

Question 1 of 36

Exam Body

Which property of real numbers is applied to the mathematical statement?

$$14 (3 + 2) = 14 (3) + 14 (2)$$

- ☐ A) Commutative Property
- ☐ B) Closure Property of Multiplication
- ☐ C) Multiplication Property of Equality
- ☐ D) Distributive Property

Question 2 of 36

Exam Body

Divide 14,218 by 212 and round your answer to the nearest hundredth. Which number will be found in the hundredth's place?

- ☐ A) 6
- ☐ B) 7
- ☐ C) 0
- ☐ D) 1

Question 3 of 36

Exam Body

The ages of the 14 teachers at Cape Fear Elementary School are listed. What is the average age of the teachers?

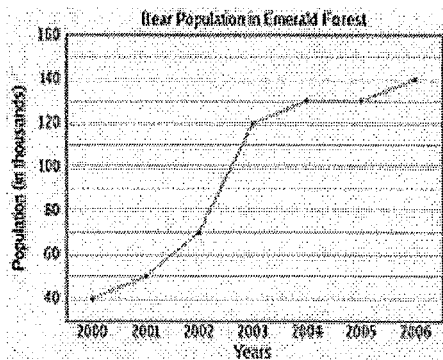
52, 36, 29, 31, 46, 28, 24, 38, 42, 28, 25, 48, 35, 42

- ☐ A) 42
- ☐ B) 29
- ☐ C) 36
- ☐ D) 41

Question 4 of 36

Exam Body

The line graph at the right gives the bear population in thousands in Emerald Forest from the years 2000 through 2006. What was the change in population in thousands from 2002 to 2005?

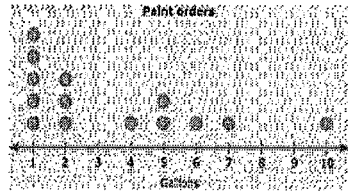


- ☐ A) 60
- ☐ B) 70
- ☐ C) 80
- ☐ D) 100

Question 5 of 36

Exam Body

The dot plot shows the number of gallons of paint ordered by customers at the Color My World paint store. What is the mode of this data?



- ☐ A) 10 gallons of paint
- ☐ B) 1 gallon of paint
- ☐ C) 5 gallons of paint
- ☐ D) 2 gallons of paint

Question 6 of 36

Exam Body

Which answer choice does not belong on a list of the multiples of 52?

- ☐ A) 936
- ☐ B) 572
- ☐ C) 1,092
- ☐ D) 754

Question 7 of 36

Exam Body

The prime factorization of two composite numbers is given. What is the prime factorization of the greatest common factor of these two numbers?

First number: $2^2 \times 3^2 \times 5^3 \times 7^2$ Second number: $2^4 \times 3^3 \times 7$

- ☐ A) $2^2 \times 3^2 \times 7$
- ☐ B) $2^4 \times 3^3 \times 5^3 \times 7^2$
- ☐ C) $2 \times 3 \times 5 \times 7$
- ☐ D) $2^2 \times 3^2 \times 5^3 \times 7^2$

Question 8 of 36

Exam Body

On Friday night, Spencer ate $\frac{1}{3}$ of the pizza he and his friends ordered. If Latitia ate $\frac{1}{4}$ of the pizza and then Mark ate all that was left, what fraction of the pizza was eaten by Mark?

- ☐ A) $\frac{1}{5}$
- ☐ B) $\frac{5}{12}$
- ☐ C) $\frac{7}{12}$
- ☐ D) $\frac{2}{3}$

Question 9 of 36

Exam Body

Maria bought $7\frac{5}{8}$ yards of fabric from a bolt of material that originally contained $18\frac{1}{3}$ yards. How many yards of fabric remained on the bolt after Maria's purchase?

- ☐ A) $11\frac{2}{3}$
- ☐ B) $10\frac{17}{24}$
- ☐ C) $12\frac{1}{8}$
- ☐ D) $8\frac{3}{8}$

Question 10 of 36

Exam Body

In a concrete mixture the ratio of bags of sand to bags of gravel is 3:4. When 8 bags of gravel are used, how many bags of sand are needed?

- ☐ A) 12 bags
- ☐ B) 10 bags
- ☐ C) 8 bags
- ☐ D) 6 bags

Question 11 of 36

Exam Body

The enrollment at Dover Elementary School is 1,225. On November 23, there were 1,127 students present. What percent of the students were absent on that day?

- ☐ A) 10%
- ☐ B) 2%
- ☐ C) 8%
- ☐ D) 6%

Question 12 of 36

Exam Body

The Miller family buys 3 quarts of milk each day. How many gallons of milk do the Millers buy in 200 days?

- ☐ A) 60
- ☐ B) 68
- ☐ C) 75
- ☐ D) 150

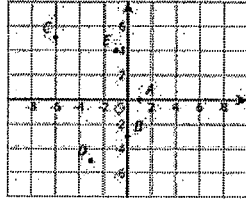
Question 13 of 36

Exam Body

Reggie works at the gym five days each week from 8:25 AM until 4:15 PM. How many hours does Reggie work in a week?

- ☐ A) 39 hours, 10 minutes
- ☐ B) 40 hours
- ☐ C) 38 hours, 10 minutes
- ☐ D) 40 hours, 50 minutes

Which point graphed on the coordinate plane lies in Quadrant III?

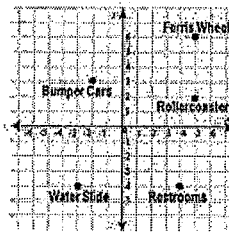


- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

Question 15 of 36

The coordinate plane shown below gives the location of some of the rides at Holiday World Amusement Park along with the location of the restrooms. Use this grid for questions 15 and 16.

What is the distance, in units, from the Ferris Wheel to the Rollercoaster?

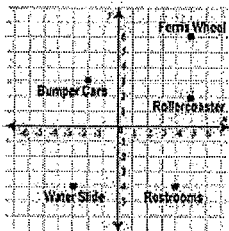


- ☐ A) 6 units
- ☐ B) 7 units
- ☐ C) 4 units
- ☐ D) 5 units

Question 16 of 36

The coordinate plane shown below gives the location of some of the rides at Holiday World Amusement Park along with the location of the restrooms. Use this grid for questions 15 and 16.

Which set of movements given will lead Sarah from the Water Slide to the Rollercoaster?

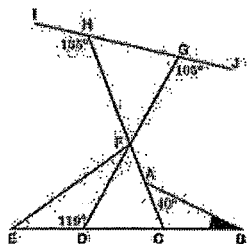


- ☐ A) 7 units left, 6 units up
- ☐ B) 8 units right, 6 units up
- ☐ C) 1 unit right, 7 units up
- ☐ D) 7 units right

Question 17 of 36

Exam Body

In the figure shown, what is the measure of $\angle FDC$?



- ☐ A) 61°
- ☐ B) 119°
- ☐ C) 45°
- ☐ D) 75°

Question 18 of 36

Exam Body

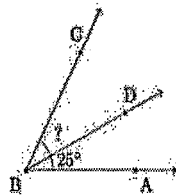
A certain four-sided convex figure has two pairs of adjacent sides that are equal in measure. It has only one pair of opposite angles that are equal. What is the specific name for this figure?

- ☐ A) Parallelogram
- ☐ B) Rhombus
- ☐ C) Kite
- ☐ D) Trapezoid

Question 19 of 36

Exam Body

In the figure shown, the measure of $\angle DBA$ is 25° . If the measure of $\angle CBA$ equals 61° , what is the measure of $\angle CBD$?



- ☐ A) 35°
- ☐ B) 25°
- ☐ C) 36°
- ☐ D) 46°

Question 20 of 36

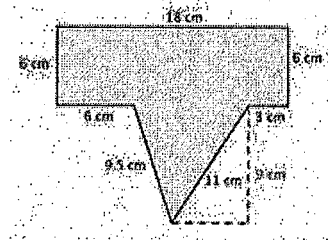
Exam Body

An urn contains 3 red marbles and 4 black marbles. Two marbles are drawn at random one at a time without replacement from the urn. What is the probability that both marbles will be black?

- ☐ A) $1/42$
- ☐ B) $2/7$
- ☐ C) $1/2$
- ☐ D) $4/7$

Use the irregular figure shown in the diagram to answer questions 21 and 22.

What is the perimeter of the irregular shape shown?

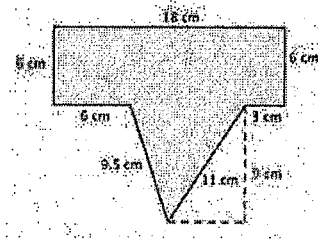


- ☐ A) 38 centimeters
- ☐ B) 44.5 centimeters
- ☐ C) 42.5 centimeters
- ☐ D) 59.5 centimeters

Question 22 of 36

Use the irregular figure shown in the diagram to answer questions 21 and 22.

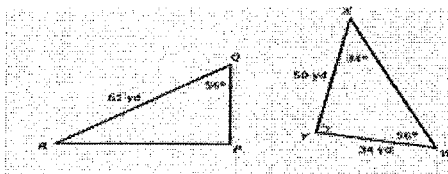
The area of the irregular shape shown is 148.5 square centimeters and represents the base of a right prism. If the height of the prism is 14 centimeters, what is the volume of this prism?



- ☐ A) 2,079 cubic centimeters
- ☐ B) 693 cubic centimeters
- ☐ C) 1,148.5 cubic centimeters
- ☐ D) 1,386 cubic centimeters

Question 23 of 36

In the figures below, $\triangle RPQ \cong \triangle XYW$. The figure is not drawn to scale. What is the length of segment RP?



- ☐ A) 61 yards
- ☐ B) 34 yards
- ☐ C) 50 yards
- ☐ D) 56 yards

Question 24 of 36

Exam Body

A standard deck of 52 playing cards contains 13 cards of each suit which are named clubs, spades, diamonds and hearts. Clubs and spades have black symbols while hearts and diamonds contain red symbols. If Jack draws one card from the deck, what is the probability it will have red symbols?

- ☐ A) $\frac{1}{4}$
- ☐ B) $\frac{1}{2}$
- ☐ C) $\frac{1}{3}$
- ☐ D) $\frac{1}{5}$

Question 25 of 36

Exam Body

What is the result of simplifying the expression? $-4 - 4x + x - 1 + 5y - 5$

- ☐ A) $-3x + 5y - 10$
- ☐ B) $2xy - 9$
- ☐ C) $-3xy - 8$
- ☐ D) $3x + 5y - 9$

Question 26 of 36

Exam Body

When the binomial $4x - 5$ is multiplied by the binomial $2x + 1$, what is the result?

- ☐ A) $8x^2 - 5$
- ☐ B) $8x^2 - 6x - 5$
- ☐ C) $6x - 4$
- ☐ D) $6x^2 - 4x - 5$

Question 27 of 36

Exam Body

What is the solution for x in the equation? $\frac{x}{2} + 5 = -2$

- ☐ A) 12
- ☐ B) -6
- ☐ C) 7
- ☐ D) -14

Question 28 of 36

Exam Body

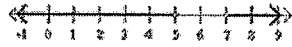
What is the value of the algebraic expression? $(5 - 1)^3 + (11 - 2)^2 - (7 - 4)^3$

- ☐ A) 118
- ☐ B) 172
- ☐ C) 192
- ☐ D) 145

Question 29 of 36

Exam Body

Which inequality statement matches the number line graph shown?



- ☐ A) $x < 5$
- ☐ B) $x \geq 7$
- ☐ C) $x < 5$ or $x > 7$
- ☐ D) $x > 5$ or $x \leq 7$

Question 30 of 36

Exam Body

Which answer choice gives one of the solutions for the equation? $x^2 - 9 = 0$

- ☐ A) $x = -9$
- ☐ B) $x = 9$
- ☐ C) $x = 3$
- ☐ D) $x = 4$

Question 31 of 36

Tiebreakers

In a standard coordinate plane, point Z is located at the ordered pair (2, -12). When point Z is reflected over the y-axis, what is the new ordered pair?

- ☐ A) (2, -12)
- ☐ B) (2, 12)
- ☐ C) (-2, 12)
- ☐ D) (-2, -12)

Question 32 of 36

Tiebreakers

Two consecutive integers have a sum of 25. What is the value of the larger of the two integers?

- ☐ A) 11
- ☐ B) 12
- ☐ C) 13
- ☐ D) 14

Question 33 of 36

Tiebreakers

What is the value of the expression when $x = -1$ and $y = -2$? $\frac{x^2 + y^2}{x^2 - y^2}$

- ☐ A) 3
- ☐ B) $\frac{4}{3}$
- ☐ C) -3
- ☐ D) $\frac{1}{2}$

Question 34 of 36

Tiebreakers

A sphere has a diameter that is 10 inches long. Using $\pi = 3.14$ and rounding to the nearest tenth, what is the volume of the sphere?

- ☐ A) 314.4 cubic inches
- ☐ B) 523.3 cubic inches
- ☐ C) 1,046.7 cubic inches
- ☐ D) 4,186.7 cubic inches

Question 35 of 36

Tiebreakers

What is the most simplified form for the algebraic expression? $3x \cdot (2x)^3$

- ☐ A) $24x^4$
- ☐ B) $6x^4$
- ☐ C) $18x^4$
- ☐ D) $24x^3$

Question 36 of 36

Tiebreakers

The scale factor of an architectural drawing of a new office building is $\frac{1}{2}$ inch : 40 feet. Each floor of the building contains a conference room with a length of 60 feet. What is the length of the conference room on the floor plan?

- ☐ A) $\frac{7}{8}$ inch
- ☐ B) $1\frac{3}{4}$ inches
- ☐ C) $1\frac{1}{2}$ inches
- ☐ D) $\frac{3}{4}$ inch

